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Benchmarking Flare Measurements Across PDO Upstream Facilities with VIIRS Nightfire Satellite Data

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Abstract

The primary objective of this paper is to analyze and validate flaring volume measurements at the oil and gas facilities of Petroleum Development Oman Company (PDO) in Oman by comparing metered flare data with annual gas flare volumes estimated by the Visible Infrared Imaging Radiometer Suite (VIIRS) Nightfire. The study evaluates the accuracy of VIIRS Nightfire data in quantifying gas flaring across PDO's wide sites, with a detailed focus on facilities exhibiting the highest data variations. Two satellite calibration methods—Cedigaz and John Zink—were applied, with John Zink demonstrated superior accuracy for small flare volumes. The study identifies factors influencing discrepancies such as data availability, synchronization issues, and metering quality. The study presents a facility-level analysis of flared gas volume estimation, focusing on the accuracy of satellite-based measurements and calibration methods. Conducted in collaboration with Petroleum Development Oman (PDO), the World Bank, and the Colorado School of Mines, the analysis covered 14 facilities and investigated discrepancies in reported flaring volumes. In Facility 'B', suspected metering inconsistencies at high pressure flare system were identified through a comparison of Cedigaz and John Zink calibration data, indicating the need for further technical investigation. For Facility 'E', deviations were primarily linked to the limitations of satellite sensors at low flow rates. A joint deep-dive with the World Bank confirmed lower detection thresholds for VIIRS sensors, prompting agreement to recalibrate flaring estimates using John Zink data. These corrections will be extended to other facilities with similar low-flow conditions.

The study also explores the influence of key variables—such as average ambient temperature, VIIRS detection frequency, and flow rate—on satellite data accuracy and the effectiveness of Nightfire (VNF) calibration. The findings underscore the importance of integrating ground-based calibration with satellite data to enhance the reliability of flared gas quantification.

Results confirm VIIRS Nightfire as an efficient verification and benchmarking measurement tool for gas flaring and support improved flare management and emissions reporting within PDO's upstream facilities, its accuracy is affected by environmental and operational limitations. In contrast, on-site ultrasonic flare meters offer more reliable and precise real-time monitoring for operational purposes. The findings

underscore the importance of integrating both approaches to enhance the overall robustness of flare monitoring and reporting frameworks.

Introduction

Gas Flare management is a core element of PDO's decarbonization strategy in line with Oman Nationally Determined Contribution (NDC) for the oil and gas sector, involving significant efforts to reduce flaring emissions and ensure accurate flare measurement and verification. PDO reports its annual total and routine flaring volumes to the World Bank under the Zero Routine Flaring Initiative; however, discrepancies have been observed between PDO's reported data and figures published by the World Bank. To address these discrepancies, a collaborative study between PDO, the World Bank, and the Colorado School of Mines (the "Study") was conducted to reconcile differences by comparing on-site measured flare volumes against VIIRS Nightfire satellite estimates across PDO operations.

The Study included PDO-wide assessment of flare measurements, focusing in-depth on facilities with the largest discrepancies. High-frequency, one-minute interval metering data for AP/LP/HP flares and flow meter readings were analyzed alongside World Bank data. Geographic coordinates were verified to ensure precise location matching. The Study also examined the impact of parameters like average temperature, VIIRS detection frequency, and flow rate on data accuracy and Nightfire calibration, aiming to enhance flare volume measurement across PDO operations.

In 2020, Oman ranked as the 10th highest country globally for gas flaring. This ranking improved to 13th place in both 2023 and 2024, according to the World Bank GFMR Global Gas Flaring Tracker Report (GFMR, June 2025). The flaring of associated gas represents a significant loss of valuable resources and contributes to environmental harm. The damage and harm referred to in this context is the environmental damage to the environment due to emitting CO₂e to the atmosphere and the equivalent monetary value lost due to gas flaring as opposed to recovering/exporting the gas which will result in an economic/monetary benefit. In 2019, flaring levels in Petroleum Development Oman (PDO) reached its peak, accounting for 28% of the company's total greenhouse gas (GHG) emissions (This is PDO's internal reference from its Scope 1 and 2 absolute GHG emissions). Since then, considerable efforts have been made to reduce this figure to around 17% of PDO's overall GHG emissions. PDO has been a participant in the World Bank's Zero Routine Flaring Initiative since 2016, committing to designing new oil fields without routine flaring and eliminating existing routine flaring by 2027, and no later than 2030. Operating within Oman's large Block 6 concession area, PDO oversees more than 100 flare stacks spread across its northern and southern regions, each with varying flow rates and gas compositions.

PDO's efforts to reduce flaring have been significant, involving major capital investments and the optimization of operational practices. Since 2020, flaring reduction has been a critical, non-negotiable target on PDO's corporate scorecard. PDO embraced the A3 strategic mindset through regular monthly flaring huddles and bi-weekly decarbonization sessions. These are regular monthly decarbonization huddles for the flaring community in PDO across assets, function and corporate teams in addition to the flaring pillar champion whose at a director level. Key initiatives include as well implementing an automated ALARP waiver system to ensure compliance, enhancing gas system reliability, repairing leaking valves, and closely tracking flaring performance as a key metric during monthly reviews. Additionally, PDO launched the "Let's Talk Flaring" campaign and integrated cultural and behavioral changes within its strengthened decarbonization strategy introduced in 2021 (Let's Talk Flaring is a campaign launched in all PDO sites operations to embrace the GHG and flaring reduction mindset and culture i.e. the operations team discuss the daily flaring performance during the daily operations meetings).

Moreover, PDO has implemented several digital initiatives to enhance flaring monitoring and performance tracking, including the creation of an Automated and Digitalized Flare Performance Dashboard. This platform offers a comprehensive breakdown of flaring activities according to World

Bank categories (Routine, Non-Routine, and Safety), presenting monthly and year-to-date flaring figures, related greenhouse gas emissions, and estimated financial losses calculated from gas flaring based on gas sales prices. Furthermore, the Energy Management Dashboard acts as a centralized hub for overseeing capital abatement projects, such as flare reduction efforts, displayed within a project maturation funnel. Users can filter these projects by asset, cluster, development stage, planned start date, and anticipated gas savings to facilitate efficient project delivery. To support its decarbonization goals, PDO also introduced an internal Energy Management & GHG online training program, emphasizing flaring reduction and fostering company-wide awareness.

PDO Flare Measurement and Verification

Flaring measurement and verification are primarily driven by regulatory requirements and environmental mandates in Oman. Worldwide, there is an increasing focus on managing flaring, methane, and carbon emissions, supported by global initiatives such as the Paris Agreement and the World Bank's Zero Routine Flaring by 2030 campaign. These efforts have contributed to measurable progress, particularly in Oman, as noted in the World Bank's June 2024 Global Gas Flaring Tracker Report.

The U.S. Environmental Protection Agency (EPA) requires flare measurement accuracy within $\pm 20\%$ for flow rates between 0.1 and 1 ft/sec, and within $\pm 5\%$ for flows above 1 ft/sec. The European Union's Emissions Trading Scheme (ETS), under Directive 2003/87/EC Annex 2, mandates accuracy levels based on annual CO_{2e} emissions: $\pm 17.5\%$ for less than 50,000 tonnes, $\pm 12.5\%$ for 50,000–500,000 tonnes, and $\pm 7.5\%$ for more than 500,000 tonnes.

PDO employs Class Four ultrasonic flowmeters for flare flow monitoring, offering $\pm 20\%$ accuracy and well-suited to the intermittent, high-temperature conditions typical of flaring. These durable meters, with an approximate 25-year lifespan, accommodate a wide range of flow rates and varying gas compositions, meeting regulatory standards.

To enhance flare measurement and verification, PDO has implemented several key initiatives aligned with global standards and best practices as follows:

- Complete mapping of flare meters across all PDO sites to ensure accurate reporting by flare stack and stream.
- Integration of flare source alerts into the Exception Based Surveillance (EBS) system for timely operational response.
- Launch of a Flare Measurement Dashboard in 2021 providing live monitoring of flare meter parameters such as pressure, flow, and temperature.
- Monthly flare huddles—Flare Gap-to-Potential and Flare Meter Health huddles—focused on improving measurement accuracy, optimizing processes, assessing project performance, and maintaining gas system reliability.
- Ongoing valve maintenance to address leaks in flare systems company-wide.
- A 2023 flare measurement campaign aligned with OGMP 2.0 commitments, helping PDO achieve the Gold Standard for methane emissions reporting in 2023 and 2024, with plans to sustain this status and reach Level 5 Gold Standard by December 2025.
- Implementation of zero drift verification protocols to maintain meter calibration accuracy over time.
- A comprehensive 2023 audit of flare measurement configurations, calibration practices, and reporting accuracy, with corrective actions tracked through PDO's Incident Management system.

Methodology, Procedure, Processes

In this paper, the study was narrowed to stations of PDO instead of focusing on the overall flaring volume. This study compared the data captured from satellites by two calibration methods (Cedigaz and John Zink) to PDO's on ground ultrasonic flare meters tracking down major discrepancies found and analyzing further.

The Global Gas Flaring Tracker Report is issued annually by the World Bank's Global Flaring Methane Reduction (GFMR) Partnership which is a group consisting of governments, oil companies and international bodies working to end routine gas flaring or Zero Routine Flaring (ZRF) at the oil production sites worldwide (GFMR, 2024). In partnership with the U.S. National Oceanic and Atmospheric Administration (NOAA) and the Payne Institute for Public Policy at the Colorado School of Mines, GFMR produces global flare estimates using satellite data from a mission launched in late 2011. This satellite is equipped with advanced sensors capable of detecting infrared heat emitted by flares at upstream oil and gas sites around the world (GFMR, 2024).

The Colorado School of Mines (CSM) in a partnership with GFMR, will receive these thermal emissions, process and analyze them using a calibration method called Cedigaz, which is an independent data provider. The flaring patterns are observed using VIIRS (Visible Infrared Imaging Radiometer Suite) sensors using the NOAA satellites. These sensors:

1. Can detect heated emissions at nighttime without interruption from moonlight or any other artificial lighting.
2. Tuned to specific wavelengths where flares will be emitting the most radiation; and
3. Ensure high-resolution imaging for precise flare identification.

Since flares burn at high temperatures, exceeding 1000°C, the VIIRS will be able to detect and pinpoint flares without minimum human input. Since launching in 2012, more than 10000 flares were identified globally per annum (GFMR, 2024).

Moreover, VIIRS sensors will use Suomi NPP and NOAA satellites for the flare stack heat emissions detection at nighttime which will enable a better estimation of the flaring volumes (CIRES, 2019). The process begins with obtaining the VIIRS NightFire data (VNF) which consist of the location of flares, the radiative power in MW and the flare thermal characteristics (CIRES, 2019). The flared gas volume will be estimated by converting the radiative power into volume taking into consideration the radiative efficiency, and the heat released rate. The formula below illustrates the flaring volume quantification using Cedigaz linear formula:

$$Volume(MMSCFD) = \frac{RP}{\eta \times HR}$$

Where RP is the Radiative Power (in MWs), η is the assumed efficiency of radiation (usually between 0.003 to 0.01) and HR is the rate of the Heat Released (typically around 50 MWs/MMSCFD).

The Cedigaz calibration assumes an efficient steady combustion of hydrocarbon of typical composition. The locations of flare stacks are verified and matched using their spatial proximity (typically $\pm 500\text{m}$) (Zhizhin, 2024). Flare observations are organized by day or month to match the timing of operational or regulatory data for comparison purposes. For more improved data accuracy, any low-confidence or outliers' detections will be removed. This method facilitates regional-scale surveillance of flaring trends, helping with regulatory compliance, emissions monitoring, and the cross-checking of reported flare volumes (Zhizhin, 2024). Furthermore, satellite estimates can achieve higher accuracy when calibrated with corresponding on-site flare meter data.

Opposing Cedigaz calibration, another calibration method has been introduced called John Zink which is considered a better and improved formula with less error. JZ applies an empirical, non-linear equation derived from controlled flare testing. The formula is:

$$Volume(MMSCFD)=\frac{A \times RP^B}{\eta}$$

The revised calibration is based on regression analysis comparing satellite-detected radiative heat with actual gas flow rates measured at flare stacks. The data used comes from 36 controlled experiments at the John Zink flare testing facility in Tulsa, Oklahoma. All tests were performed under standardized conditions, clear skies, minimal wind and consistent fuel composition. During the experiments, specific variables were intentionally varied, including gas flow rates (from 750 to 75,000 lb/hr), satellite viewing angles (from directly overhead to 70° side), and the number of active flare stacks (either one or two burning simultaneously) (Zhizhin, 2024).

Table 1—Comparison Summary between Cedigaz and John Zink

Calibration Type	Cedigaz	John Zink
Basis	Thermodynamic/Energy balance approach	Empirical model derived from 36 controlled flare tests
Assumptions	Constant radiative efficiency and heat released rate	Efficiency and relationship derived from experimental regression of Radiative Power vs flowrate
Strengths	Simple & widely applicable – usually global, top-down monitoring	More accurate for site-specific conditions.
Limitations	Overestimates small flares and underestimate large flares	Requires detailed test data and less straightforward for large-scale global use.
Applications	Used for Global Monitoring (e.g. GFMRM, WB, NOAA...)	Engineering Analysis, flare technology design.
Best suited for	Large scale/Global estimates – where site related data are unavailable.	Local/site calibration where both flare and fuel data are known.

The analysis of test data resulted in a simplified calibration linking radiative heat (RH) directly to flared gas volume (Q) through a linear relationship, removing the need for flare shape correction factors. However, for medium to large flares, the VIIRS Nightfire (VNF) system often detected flare activity across multiple adjacent pixels in the infrared imagery. To enhance accuracy, RH values from nearby pixels were included, provided each pixel's RH was at least 75% of the maximum RH within the detected cluster (Zhizhin, 2024). This method notably strengthened the correlation between actual metered gas flow and satellite-derived RH, leading to the development of the following calibration equation:

$$Q_{JZ}(BCM)=0.012 \times RH_{75\%}(MW)$$

The accuracy of VIIRS Nightfire in estimating flared gas volumes at two Oman flare locations was evaluated using both the Cedigaz and John Zink calibration models. For compatibility with on-site measurements, flow rates were converted to million standard cubic meters per day (Sm³/day), instead of being presented in billion cubic meters per year (bcm/y)

Figure 1 illustrates the variations between the two calibration methods, using flow rate estimates from all global flares detected by VIIRS on 15/08/2024, as documented in the multiyear database. The nonlinear behavior of the Cedigaz model generally leads to overestimation of smaller flare volumes (Q < 0.5 Mm³/day) and underestimation of larger ones, while the John Zink model delivers more consistent and balanced estimates across different flare sizes (Zhizhin, 2024).

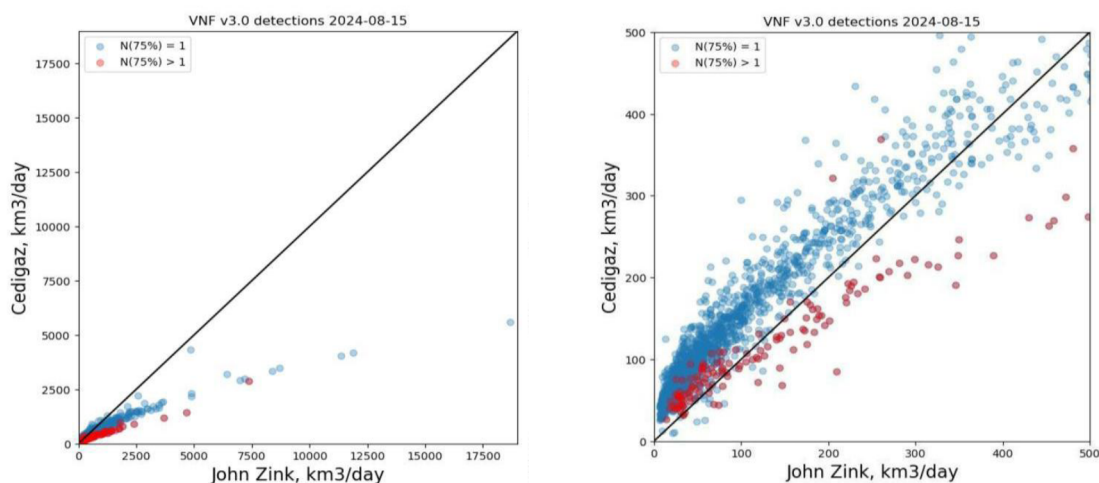


Figure 1—Cedigaz & JZ comparison by VNF on 15/08/2024. Left: full-scale comparison. Right: Close-up analysis of flares emitting less than 500 km³/day

Relative to the satellite measurement, it is required to have meters on ground for the comparison to happen and in PDO, ultrasonic flow meters are used. This procedure is essential for comparing satellite observations with on-site measurements. Ultrasonic meters send all collected data to the Distributed Control System (DCS), which tracks both primary readings and associated instrumentation. Flow measurements are compensated using the AGA-10 international standard, integrated within the DCS, accounting for variables such as actual flow, pressure, temperature, and gas compressibility. The calculated output is sent to the current totalizer, which logs the cumulative volume. The DCS is also connected to additional software systems, including Process Information (PI), Nibras, and the Production Measurement Management System (PMMS), enabling continuous monitoring, trending, and analysis of both actual and standardized data.

Results and Observation

The World Bank flare tracking tool is an independent indicator of flared gas around the globe, the tool monitors global flared volumes with a motive to showcase global transparency, environmental responsibility and economic waste to reduce one of the oil and gas most visible forms of environmental waste, this helps to drive the change and progress toward zero routine flaring by 2030.

Emission reporting in World Bank flaring figures is predominantly utilizing satellite-based techniques to capture the temperature and hence calculates flared volume, while on the other hand oil and gas companies invest on-site meters to capture real time flared volumes considering the industry practice of meter maintainability and accuracy limits.

When designing a flare header, it is essential to include a measurement device downstream of the knock-out drum. International standards provide a wide range of selection criteria for flare meters, primarily based on the required measurement accuracy and the balance between capital and operational expenditure. Considering these factors, many oil and gas operators recommend ultrasonic flowmeters, as they are capable of measuring large flow ranges with high accuracy while requiring minimal maintenance.

However, certain design requirements must be addressed when installing an ultrasonic flowmeter. These include ensuring minimum straight-pipe lengths upstream and downstream of the meter, as well as installing pressure and temperature transmitters. The transmitters are necessary to convert the actual flow to standard conditions, typically defined at a base pressure of 101.325 kPa and a base temperature of 15 °C. Standardizing the volume in this way reduces measurement uncertainty by accounting for variations in flow due to pressure and temperature changes, thereby improving the accuracy of the reported flare gas volumes.

Satellite data, on the other hand, uses a fundamentally different approach to estimate flare volumes. Accurate readings require proper mapping of the flare location based on latitude and altitude to measure the heat radiation detected from the ground. A key limitation is that satellites cannot reliably distinguish between flare stacks located within a few hundred meters of each other. In this study, many facilities had multiple flare stacks in close proximity—for example, Atmospheric Pressure (AP) and Low Pressure (LP) flares separated by only a few hundred meters. In such cases, the satellite system recorded the combined heat signature and reported a total flare volume rather than separating the contributions of each stack.

Another challenge with satellite measurements is environmental interference. Cloud cover and other weather conditions can partially or fully block the heat signal, resulting in missing or unreliable data. Furthermore, satellites typically capture only a single measurement for each location per day, with the result recorded as a daily average of flared volume until the next pass.

The main drive of comparing flowmeters on ground with satellite data is the high discrepancy on volumes between both in some locations within PDO, the difference is illustrated on the below [Figure 2](#) for 2022 as a comparison between satellite and meters in kSm³/d, 2023 [Figure 3](#) has been accommodated to have a more comprehensive results and fair comparison.

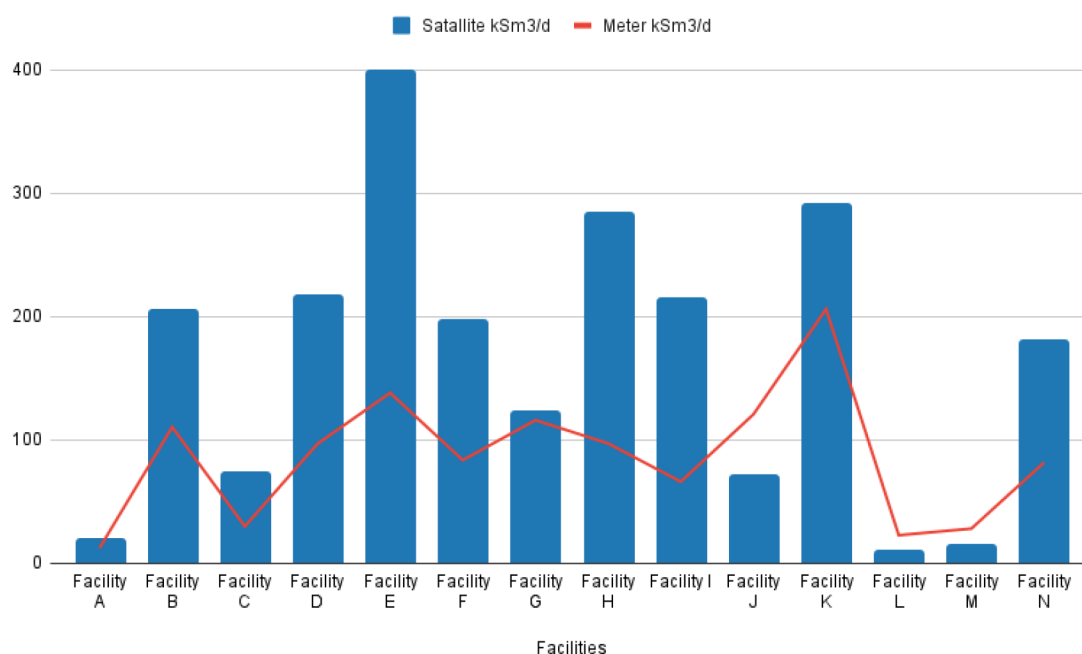


Figure 2—2022 flaring data comparison between satellite and meters in kSm³

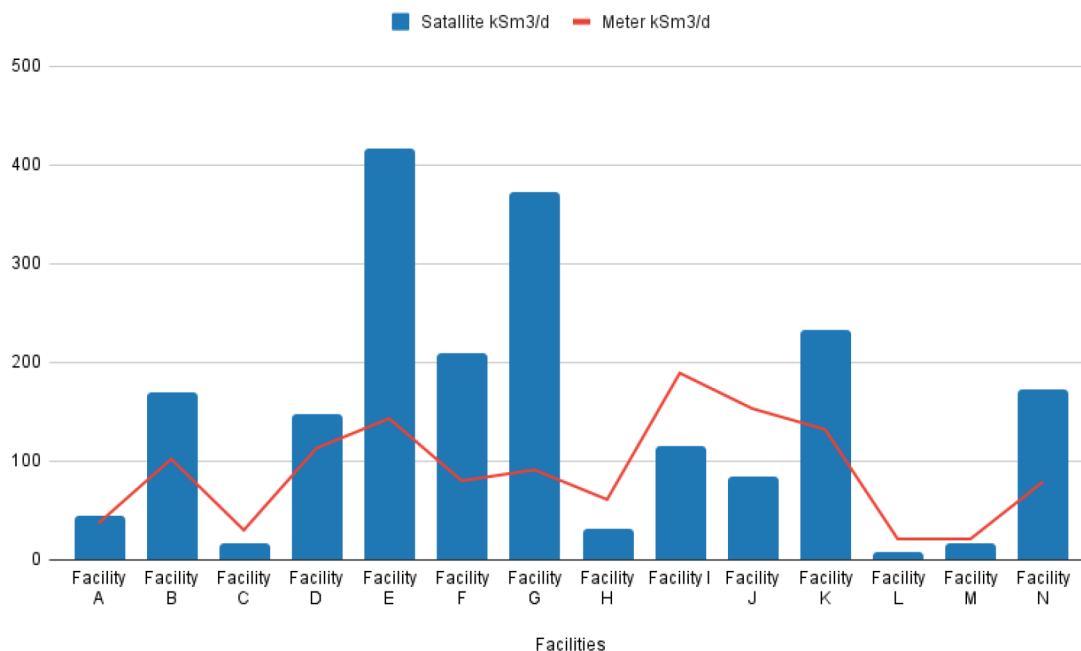


Figure 3—2023 flaring data comparison between satellite and meters in kSm3

The data above represent the yearly average of few selected facilities in PDO compared to satellite readings recorded in the world bank, it is worth noting that each facility consist of multiple flare stacks though since the satellite couldn't differentiate between multiple stacks within the same facility, flare meter data was summed up for each facility to have a fair comparison with satellite readings, in most of the facilities the satellite was over reading the on-site flare meters while in few others it was the opposite.

To understand the discrepancy further, zooming in to a specific facility unleashes more insights on the behavior of satellite readings versus flare meters, considering facility E daily averaged figures compared to flare meters readings at the same time span represented in Figure 4, the satellite data shows a high variance and noise multiple times more than the flare meter which showed a stable reading and below a specific threshold aligned with the operation philosophy of the facility. Throughout end of the year the satellite reading declined dramatically lower than the actual meters though the satellite have recorded heavy clouds blocking the sight of the facility flare heat source.

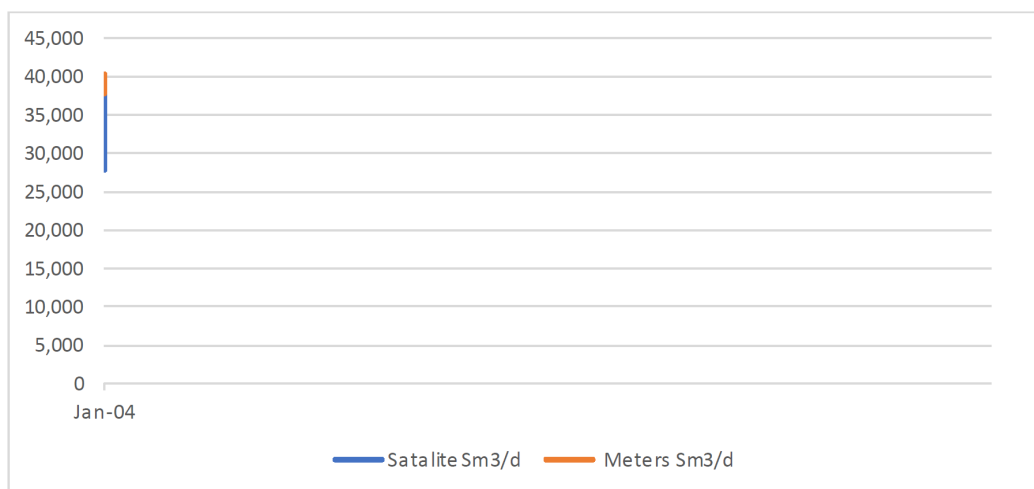


Figure 4—Facility E flare data comparison between flare meters and satellite

facility B was also trended in a daily time span to further analyze the discrepancies associated with the satellite versus the flow meter readings, this specific facility had a high pressure flare system which is used to depressurize the system of gas during any process upsets or emergency blowdowns hence a high flare flow rates are expected, as per Figure 5 many spikes are noted due to the event of blowdown and it has been realized that the flowmeter maximum range was set at 650KSm³/d the meter has reached the maximum few times in the year though the satellite has over read the meter twice times the reading of the meter and the capacity of the facility, unsurprisingly facility W and facility E are within the same area thus the cloud season affected both facilities, though in facility B the satellite have shown close readings with the flare meters with a deviation of approximately 5% difference.

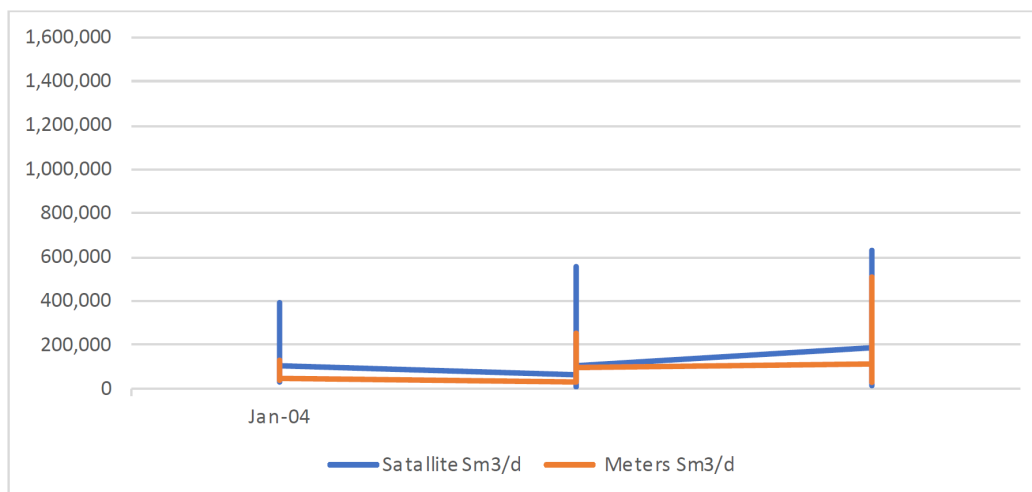


Figure 5—Facility B flare data comparison between flare meters and satellite

The Study has covered other facilities to land into a common ground, the satellite reading had some back-end limitation to record flares less than 10 Sm³/d as the heat source cannot be captured, many instances found the satellite over read the flare meters with a variance of more than 10% at such low flow scenarios hence those facilities were excluded due to high uncertainty of satellite readings.

Conclusion

This study evaluated flaring emissions at 14 PDO facilities by comparing on-site ultrasonic flare meter measurements with satellite-based estimates derived using the Cedigaz and John Zink calibration methods applied to VIIRS NightFire data. The comparison aimed to quantify discrepancies and assess the factors influencing measurement accuracy.

Detailed analysis of individual facilities provided deeper insights into the differences between the two approaches. At Facility E, daily averaged data revealed that satellite readings exhibited significantly higher variance and noise compared to the stable readings from the flare meters, which remained within operational thresholds. Towards the end of the year, satellite readings dropped well below actual metered values, coinciding with periods of heavy cloud cover that obscured the flare heat source.

At Facility B, which operates a high-pressure flare system for depressurizing gas during process upsets or emergency blowdowns, high flare flow rates and spikes were expected. The on-site meter, with a maximum range of 650 kSm³/d, reached its limit several times, while satellite readings occasionally exceeded both the meter output and the facility's physical capacity. Despite this, Facility B showed close agreement between satellite and metered data, with an average deviation of around 5%. Facilities W and E, located in the same area, were similarly impacted by seasonal cloud cover.

Across the wider dataset, satellite readings demonstrated limitations in detecting flares below 10 Sm³/d, often overestimating meter values by more than 10% in low-flow scenarios. Due to the high uncertainty in such cases, these facilities were excluded from the final analysis.

Overall, the findings highlight that while satellite-based monitoring offers valuable independent verification and broad coverage, it is subject to environmental and operational constraints that can impact accuracy. On-site ultrasonic flare meters remain the more reliable source for real-time operational monitoring, while satellite data serves best as a complementary tool for transparency, benchmarking, and long-term trend analysis.

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